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Studierichting: Informatica
Oefeningenreeks 2D nr. 6.6

$$f(x) = \frac{x}{4x+1}$$

geval $x \to 0$:

$$\lim_{x\to 0} \frac{x}{4x+1}$$
$$= \frac{0}{4 \cdot 0 + 1}$$
$$= \frac{0}{1}$$
$$= 0$$

geval $x \to -1$:

$$\lim_{x\to -1} \frac{x}{4x+1}$$
$$= \frac{-1}{4 \cdot (-1) + 1}$$
$$= \frac{-1}{-4+1}$$
$$= \frac{-1}{-3}$$
$$= \frac{1}{3}$$

geval $x \to 1$:

$$\lim_{x\to 1} \frac{x}{4x+1}$$
$$= \frac{1}{4 \cdot 1 + 1}$$
$$= \frac{1}{4+1}$$
$$= \frac{1}{5}$$

geval $x \to \frac{-2}{3}$:

$$\lim_{x\to \frac{-2}{3}} \frac{x}{4x+1}$$
$$= \frac{\frac{-2}{3}}{4\frac{-2}{3} + 1}$$

$$
\begin{aligned}
&= \frac{\frac{-2}{3}}{\frac{-8}{3}+\frac{3}{3}} \\
&= \frac{\frac{-2}{\cancel{3}}}{\frac{-5}{\cancel{3}}} \\
&= \frac{-2}{-5} \\
&= \frac{2}{5}
\end{aligned}
$$

References